

Amirreza Sokhankhosh

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HIGHLIGHT OF QUALIFICATIONS

- **Computer Vision & Deep Learning:** Experienced in building and optimizing **CNNs**, **Vision Transformers (ViT)**, and **Diffusion Models** using **PyTorch** and **TensorFlow** for tasks such as detection, segmentation, and generative image synthesis.
- **Model Evaluation & Research:** Designed advanced evaluation pipelines for generative models, developing the **Unified Image Evaluation Metric (UIEM)** that integrates fidelity, diversity, and generalization metrics.
- **Applied ML Engineering:** Developed scalable, containerized **ML pipelines** for distributed training and real-time inference, leveraging **Docker**, **REST APIs**, and **microservice architecture** to ensure modularity.
- **Image Processing & Data Engineering:** Skilled in **OpenCV**, **NumPy**, and **Pandas** for preprocessing, augmentation, and feature extraction, ensuring robust model generalization in real-world computer vision pipelines.
- **Software Development:** Strong foundation in **backend systems and APIs** using **Flask**, **Django**, and **Express**, and full-stack integration with **React (TypeScript)** for deploying end-to-end vision-based applications.
- **Cloud & Experiment Tracking:** Hands-on with **AWS**, **Docker**, **TensorBoard**, and **Weights & Biases** for experiment management, CI/CD, and scalable deployment of trained models.

EXPERIENCES

University of Manitoba

ML Engineer (Contract)

May 2025 – Present

Winnipeg, Canada

- Preprocessed and normalized large **IoT datasets** in, producing model-ready inputs that stabilized convergence.
- Built a scalable ML pipeline with **PyTorch**, enabling secure, **parallel training** across 8+ distributed nodes with zero raw data exposure.
- Designed and deployed a Temporal Fusion Transformer for **multi-variate time-series forecasting**, reducing prediction errors by >20% for temperature, CO₂, and humidity data.
- Engineered a blockchain-integrated backend using Hyperledger Fabric and Express.js to automate model aggregation for 76 IoT sensor streams.

University of Manitoba

Graduate Research Assistant

Sep 2023 – Jul 2025

Winnipeg, Canada

- Designed **4** end-to-end distributed **ML pipelines** with **PyTorch** and **TensorFlow**, using modular code and containerization, supporting up to **36** training nodes.
- Built a microservice-based orchestration layer with **Flask** and **Express.js** and a fast **REST API** to coordinate distributed training reliably, achieving **90–100%** node uptime across all runs.
- Improved **model robustness** via adaptive validation schemes, handling inconsistent or adversarial updates, achieving **100%** detection of simulated adversaries.
- Optimized **training efficiency** by offloading heavy computation to servers, enabling lightweight clients, and reducing iteration time from **hours to minutes**.
- Co-authored **5+** **research papers** in distributed ML engineering, published/submitted to **top IEEE venues**.

Bobo App Ltd.

Full Stack Developer Intern

May 2024 – Aug 2024

Winnipeg, Canada

- Faced the challenge of improving customer support efficiency; collaborated with a team at **MILA** to develop an **intelligent chatbot** that automated routine customer interactions, reducing response times.
- Accelerated backend development by designing and implementing **RESTful APIs** with **PostgreSQL**, improving data retrieval performance and supporting scalable product features.

- Streamlined data management by automating CSV-to-SQL integration with Python scripts, reducing manual database population time by **80%**.
- Ensured smooth **full-stack integration** through cross-functional collaboration with the front-end team, aligning API contracts with user interface requirements to enhance product reliability.
- Enhanced team productivity within an **agile workflow** by using **Jira** and **Confluence** for task tracking, documentation, and sprint coordination.

Nadin Soft (Sadr Group Company)

Jul 2020 – Dec 2020

Full-stack Developer Intern

- Drove the successful completion of a critical internal project, addressing key deficiencies in API security, authentication, and data modeling for production deployment.
- Developed robust RESTful APIs with **Express.js**, implementing **JWT authentication** to secure user access and enable core application functionalities.
- Designed and implemented highly accurate MySQL database schemas for precise tracking and management of employee working and non-working hours, foundational for the project's business logic.
- **Deployed** fully functional features to the **live production server**.
- Contributed to an Agile Scrum development team, ensuring efficient progress and clear communication within sprints and **CI/CD pipelines**.

Image Generative Model Evaluation

May 2024 – Jun 2024

Personal Project | [Code](#)

- Developed an end-to-end **evaluation pipeline** for diffusion generative models using **PyTorch** and **TorchVision**.
- Integrated a fine-tuned **ResNet-50** feature extractor to compute embeddings from real and generated images.
- Implemented custom **evaluation metrics**—Precision, Recall, Generalization Rate, and F1 Score—using nearest-neighbor searches with **scikit-learn**, and designed a unified framework combining sample fidelity and diversity into a single **Unified Image Evaluation Metric (UIEM)** score.

Facial Boundary & Landmark Detection

May 2022 – Jun 2022

Personal Project | [Code](#)

- Developed a **two-stage deep learning pipeline** combining **R-CNN** for facial boundary detection and a custom **CNN** for 68-point facial landmark prediction using **TensorFlow/Keras**.
- Implemented transfer learning with a **VGG16 backbone** for high-precision face detection and optimized model performance through fine-tuning and custom IoU-based loss functions.
- Integrated preprocessing and augmentation pipelines with **OpenCV**, **NumPy**, and **Pandas** to improve model generalization and real-world inference reliability.

EDUCATION

University of Manitoba

Sep 2023 – Aug 2025

Master of Science in Computer Science (GPA: 4.4 / 4.5)

Winnipeg, Canada

- **Relevant Coursework:** Security & Privacy, Deep Generative Modeling, Data Mining: A+

K.N. Toosi University of Technology

Sep 2018 – Feb 2023

Bachelor of Science in Computer Engineering

TECHNICAL SKILLS

Computer Vision: OpenCV, PyTorch, TensorFlow, Detectron2, TorchVision

Deep Learning: CNNs, Vision Transformers (ViT), Autoencoders, GANs, Diffusion Models

Data Analysis & Visualization: Pandas, Matplotlib, Seaborn, Plotly

Programming Languages: Python, C++, Java, JavaScript, TypeScript

Web & API Development: *Back-end:* Flask, Django, Express; *Front-end:* React, Material UI

Databases: PostgreSQL, MongoDB, MySQL, Redis

Cloud & DevOps: Docker, Linux, AWS, CI/CD, Git/GitHub

Tools & Methodologies: TensorBoard, Weights & Biases, Jira, Agile, Scrum